



ASSESSMENT COVERSHEET

Trimester 2620

Attach this coversheet as the cover of your submission. All sections must be completed.

Section A: Submission Details

Programme : Bachelor of Computer Science
Course Code & Name : LDCW6123 - FUNDAMENTALS OF DIGITAL COMPETENCE FOR PROGRAMMER
Course Lecturer(s) : Mr. V Segaran A/L Veeraya
Submission Title : Group 13 Spotify and Physical Music Clayton Model and Interactive Program
Deadline : Day 02 Month 10 Year 2026 Time 23:59
Penalties :

- Plagiarised work is an Academic Offence in University Rules & Regulations and will be penalised accordingly.
- Failure to fulfil the limits set for plagiarism similarity index and AI-generated content will result in the **essay being awarded a score not exceeding 50%** of the total score.
- Without Turnitin reports submission is deemed **incomplete**.

Section B: Academic Integrity

Tick (✓) each box below if you agree:

- | | |
|--------------------------|--|
| ☐ | I have read and understood the policy on Plagiarism in University Rules & Regulations. |
| ☐ | This submission is my own, unless indicated with proper referencing. |
| ☐ | This submission has not been previously submitted or published. |
| ☐ | This submission follows the requirements stated in the course. |

Section C: Submission Details (Group 13)

Student ID(s)	Student Name(s)	Designation
261UC240LY	Aziel Tan Zheng Chuan	Group leader
261UC240PJ	See Wing Kit	Member
261UC26145	Soo Kian Rong	Member
261UC2406W	Vincent Lock Chun Kit	Member
253UC255P1	Wong Kee Yuan	Member
262UC250T8	Ho Ming Hao	Member

Declaration by group leader

I hereby declare that all group members' names are correctly included in the above section. I hold a copy of this assignment which I can produce if the original is lost or damaged. I certify that no part of this assignment has been copied from any other student's work or from any other source except where due acknowledgement is made in the assignment.

Group leader's signature : _____

Group leader's name: Aziel Tan Zheng Chuan

Group leader's student ID: 261UC240LY

Date : _____

Group member declaration

(Each group member, including the **group leader**, must individually fill up and submit this form)

For the purpose of completing this assignment, I have performed the following tasks (please list):

I hereby declare that I have assessed this submission and I take full responsibility should there be any inaccuracies, incompleteness, omissions, delays or non-submission.

Group member's signature : _____

Group member's name: Aziel Tan Zheng Chuan

Group member's ID: 261UC240LY

Date : _____

Group member declaration

(Each group member, including the **group leader**, must individually fill up and submit this form)

For the purpose of completing this assignment, I have performed the following tasks (please list):

I hereby declare that I have assessed this submission and I take full responsibility should there be any inaccuracies, incompleteness, omissions, delays or non-submission.

Group member's signature : _____

Group member's name: See Wing Kit

Group member's ID: 261UC240PJ

Date : _____

Group member declaration

(Each group member, including the **group leader**, must individually fill up and submit this form)

For the purpose of completing this assignment, I have performed the following tasks (please list):

I hereby declare that I have assessed this submission and I take full responsibility should there be any inaccuracies, incompleteness, omissions, delays or non-submission.

Group member's signature : _____

Group member's name: Soo Kian Rong

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Date : _____

Group member declaration

(Each group member, including the **group leader**, must individually fill up and submit this form)

For the purpose of completing this assignment, I have performed the following tasks (please list):

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Group member's signature : _____

Group member's name: Vincent Lock Chun Kit

Group member's ID: 261UC2406W

Date : _____

Group member declaration

(Each group member, including the **group leader**, must individually fill up and submit this form)

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Group member's signature : _____

Group member's name: Wong Kee Yuan

Group member's ID: 253UC255P1

Date : _____

Group member declaration

(Each group member, including the **group leader**, must individually fill up and submit this form)

For the purpose of completing this assignment, I have performed the following tasks (please list):

I hereby declare that I have assessed this submission and I take full responsibility should there be any inaccuracies, incompleteness, omissions, delays or non-submission.

Group member's signature : _____

Group member's name: Ho Ming Hao

Group member's ID: 262UC250T8

Date : _____

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Spotify and Physical Music Clayton Model and Music Discovery Assistant

LDCW6123 | Trimester 2620 | Group 13 | Class section: [CONFIRM CLASS SECTION]

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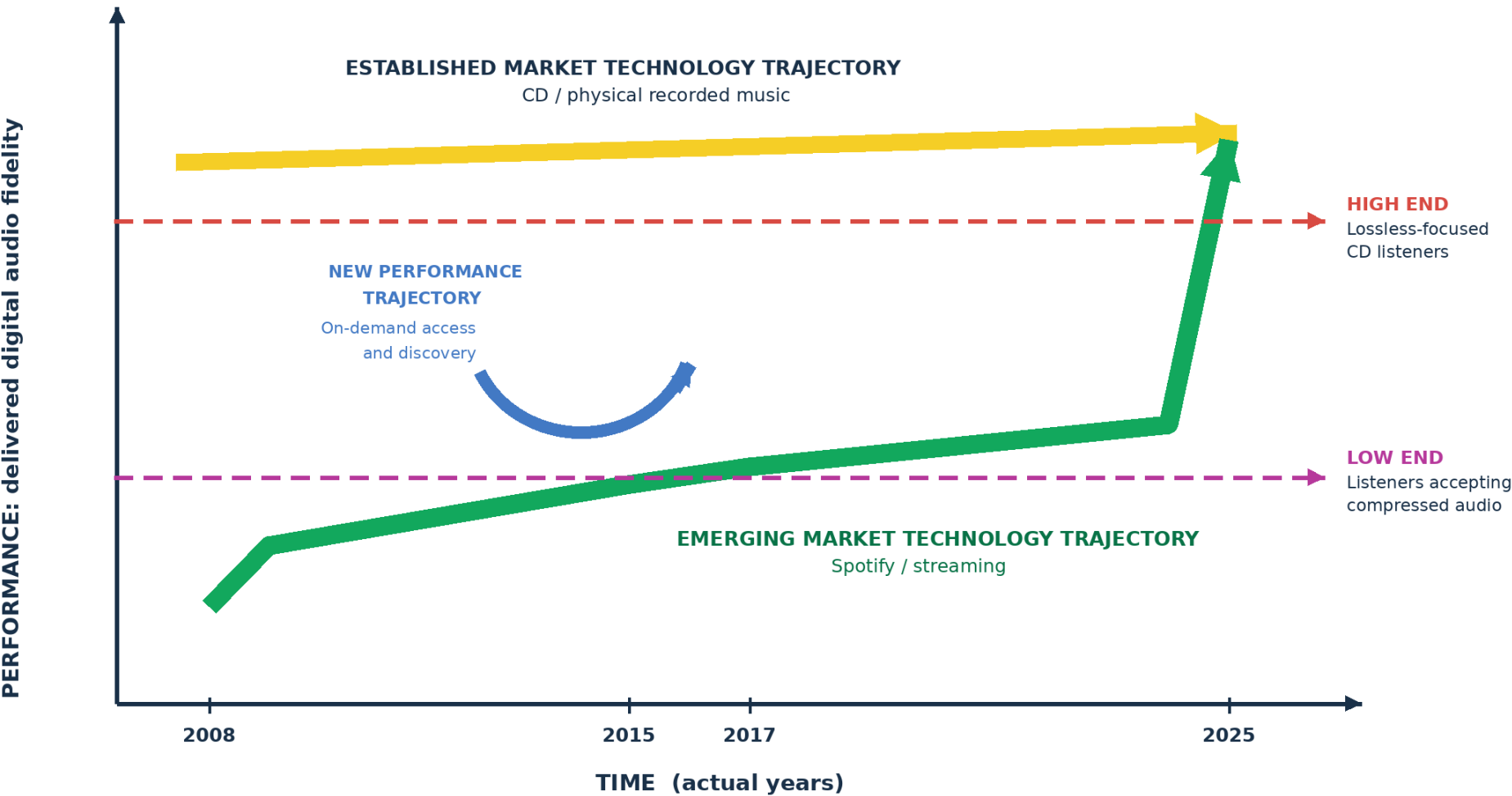
Part 1 applies the lecturer's Clayton graph to CD listening and Spotify streaming with one stated audio-fidelity metric. Part 2 shows an offline, rule-based music-discovery demonstration. Historical events and a conceptual model interpretation are kept distinct.

FROM OWNERSHIP TO ACCESS

Spotify vs CDs / Clayton Christensen disruptive innovation model / LDCW6123 Group 13

SPOTIFY vs CDs | CLAYTON DISRUPTIVE MODEL

Your technology: Spotify on-demand music streaming



2008 launch | 2009 mobile | 2015 discovery | 2017 industry streaming revenue lead | 2025 Premium Lossless
Conceptual placement only. 2017 revenue is an industry outcome, not audio parity; CD format remains fixed.

ESTABLISHED MARKET

A purchased CD provides a lossless 16-bit/44.1 kHz digital signal and lasting possession of that disc (Sony, n.d.).

EMERGING ENTRANT

Spotify launched in 2008 with access to licensed music. Early compressed streaming traded conventional signal fidelity for immediacy (Spotify, n.d.).

DIFFERENT VALUE

Search and discover music on demand without buying each CD. Mobile access arrived in 2009; Discover Weekly followed in 2015 (Spotify, n.d.).

CUSTOMERS

High-end: lossless-focused CD listeners. Lower conventional demand: people willing to accept compressed playback for access. These are needs, not income groups.

MODEL LIMIT

The near-level CD path reflects a fixed format. The 2017 industry revenue lead is not an audio-quality crossing or proof that Spotify alone caused CD decline.

2008

Spotify launches in six European markets

2009

Mobile access

2015

Discover Weekly

2017

Streaming leads global recorded-music revenue*

2025

Premium Lossless rollout; physical formats persist

SELECTED REFERENCES

Christensen Institute. (n.d.). Disruptive innovation theory.
<https://www.christenseninstitute.org/theory/disruptive-innovation/>

International Federation of the Phonographic Industry. (2018). IFPI global music report 2018.
<https://www.ifpi.org/ifpi-global-music-report-2018/>

International Federation of the Phonographic Industry. (2026, March 18). Global music report 2026: Global recorded music revenues grow 6.4% as record companies drive innovation.
<https://www.ifpi.org/global-music-report-2026-global-recorded-music-revenues-grow-6-4-as-record-companies-drive-innovation/>

Sony. (n.d.). A quick guide in understanding digital audio.
<https://www.sony.com/electronics/support/articles/00165079>

Spotify. (2025, September 10). Lossless listening arrives on Spotify Premium with a richer, more detailed listening experience.
<https://newsroom.spotify.com/2025-09-10/lossless-listening-arrives-on-spotify-premium-with-a-richer-more-detailed-listening-experience/>

Spotify. (n.d.). Spotify: A visual history.
<https://newsroom.spotify.com/spotify-timeline/>

*IFPI (2018) reports 38.4% of global industry revenue in 2017. Conceptual graph: vertical positions are not numerical measurements.

Part 1 Model application

Scope and model

The incumbent is purchased compact discs (CDs) within the physical recorded-music market. The entrant is Spotify's licensed on-demand streaming service, launched in six European countries in October 2008 (Spotify, n.d.). Christensen's model asks whether an entrant first satisfies less demanding or overlooked users on a conventional performance dimension while offering another form of value, then becomes more capable over time (Christensen Institute, n.d.). This poster applies that test to streaming; it does not equate any new digital service with a proven low-end disruption.

Foothold and customers

A CD listener seeking an owned, lossless digital recording has a higher fidelity requirement. A less demanding listener may accept compressed streaming when convenient search and access to many licensed tracks matter more than ownership or maximum fidelity. "Low-end" here refers to the tolerance for a lower level on this particular metric, not to a person's income. Spotify's free and subscription options illustrate a different way to access music, although the early market also included other digital alternatives.

Performance and customer mapping

The graph's single vertical metric is the fidelity of the delivered digital music signal relative to a lossless source under suitable playback conditions. Sony identifies standard CD audio as 16-bit/44.1 kHz. Earlier Spotify streaming used compressed audio; Spotify announced a Premium lossless option with up to 24-bit/44.1 kHz FLAC in September 2025 (Sony, n.d.; Spotify, 2025). The yellow CD path is almost level because the CD format did not continuously gain bit depth. The green path begins lower, crosses a conceptual good-enough threshold for compression-tolerant listeners and ends at a supported lossless option. Arrow slopes and threshold crossings are schematic, not measured scores or dated parity observations.

A second, distinct value

The template's blue curved callout is not another vertical scale. It denotes Spotify's alternative attraction: on-demand catalogue access, instant search and later discovery tools such as Discover Weekly, which Spotify dates to 2015 (Spotify, n.d.). Listening without owning each disc reduced a purchase and storage barrier. This benefit explains why a user might choose an entrant whose conventional audio fidelity was initially lower.

Technology and business model

Licensing and digital distribution allowed one service to supply on-demand tracks across devices; mobile access followed in 2009, according to Spotify's chronology. The free tier and paid Premium offered different access and quality options. The commercial arrangement differs from selling a permanently owned CD. These differences connect the technology, the business model and the audience's trade-off (Spotify, n.d.; Christensen Institute, n.d.).

Part 1 Historical development and impact

Verified timeline

Spotify launched in October 2008, went mobile in September 2009, introduced a free mobile experience in 2013 and launched Discover Weekly in 2015 (Spotify, n.d.). IFPI reported that streaming became the largest single source of global recorded-music revenue in 2017, accounting for 38.4% of the total. That figure combines streaming services and is not a Spotify market-share statistic (IFPI, 2018). Spotify began rolling out Premium Lossless in September 2025 in selected markets (Spotify, 2025).

Interpreting the performance path

Early compressed playback was below a CD's lossless digital format on this narrowly defined metric. The conceptual lower demand line denotes listeners for whom streaming quality was adequate even before lossless parity. By 2017, streaming as a format led global recorded-music revenue despite the absence of a Spotify lossless tier; therefore its market success cannot be explained by audio parity alone. The green endpoint marks a later improvement available to eligible Premium listeners, not a claim that every stream, device or track is lossless (IFPI, 2018; Spotify, 2025).

Access, revenue and social impact

Digital access shifted listening from purchasing specific physical albums towards searching a licensed catalogue and paying through subscriptions or advertising. Portability, recommendations and immediate discovery could benefit listeners with little storage or an inconvenient retail option. IFPI reported total streaming at 69.6% of global recorded-music revenue in 2025, while physical formats also grew by 8.0%. These are industry-wide format outcomes, not evidence that Spotify alone caused CD decline or that physical media vanished (IFPI, 2026).

Part 1 Tradeoffs and critical interpretation

Costs and limits

A purchased CD can be kept and played with compatible equipment without renewing a streaming subscription; licensed streaming is tied to available tracks, devices and service terms. Spotify Premium offers downloads for offline listening, but this still differs from owning a disc (Spotify, 2019). The quality comparison depends on encoding, plan, device and output path; Spotify specifically cautions that Bluetooth transmission may compress a lossless source (Spotify, 2025). Vinyl collectors and other physical-music buyers remain part of the market (IFPI, 2026).

Critical interpretation

Spotify is a useful illustration of a lower conventional performance trade-off and a new access value, but a strict Christensen classification is contestable. The incumbent record companies also licensed streaming, earlier digital downloads and other services played roles, and 2017 streaming revenue does not prove the model's mechanism. The graph is a reasoned schematic, not causal identification, fidelity test results or a market-share plot. The CD line is nearly level to preserve the real format ceiling while retaining the lecturer's upper incumbent trajectory.

Connection to Part 2

The offline C++ Music Discovery Assistant illustrates just one part of the new value proposition: searching a small catalogue by stated preferences. It uses fictional artists and transparent rules. It neither uses Spotify accounts or data nor reproduces Spotify's recommendation algorithms.

Part 1 References

Christensen Institute. (n.d.). *Disruptive innovation theory*.

<https://www.christenseninstitute.org/theory/disruptive-innovation/>

International Federation of the Phonographic Industry. (2018). *IFPI global music report 2018*.

<https://www.ifpi.org/ifpi-global-music-report-2018/>

International Federation of the Phonographic Industry. (2026, March 18). *Global music report 2026:*

Global recorded music revenues grow 6.4% as record companies drive innovation.

<https://www.ifpi.org/global-music-report-2026-global-recorded-music-revenues-grow-6-4-as-record-companies-drive-innovation/>

Multimedia University. (n.d.). *Digital convergence and emerging innovative problem solving through:*

Topic 6A [Lecture slides]. LDCW6123.

Part 1 References continued

Sony. (n.d.). *A quick guide in understanding digital audio.*

<https://www.sony.com/electronics/support/articles/00165079>

Spotify. (2019, May 24). *How to download and listen to music and podcasts - offline and on the go.*

<https://newsroom.spotify.com/2019-05-24/how-to-download-and-listen-to-music-and-podcasts-offline-and-on-the-go/>

Spotify. (2025, September 10). *Lossless listening arrives on Spotify Premium with a richer, more detailed listening experience.*

<https://newsroom.spotify.com/2025-09-10/lossless-listening-arrives-on-spotify-premium-with-a-richer-more-detailed-listening-experience/>

Spotify. (n.d.). *Spotify: A visual history.*

<https://newsroom.spotify.com/spotify-timeline/>

Part 2 Programme design

Purpose and integration

The Music Discovery Assistant lets a user browse 12 fictional tracks by genre, mood and preferred duration. It demonstrates the catalogue-discovery aspect of access-based music services without connecting to Spotify or an external API.

Inputs and outputs

Genres are Pop, Electronic and Acoustic. Moods are Calm, Energetic and Reflective. A length option asks for up to four minutes or over four minutes. Output shows the fictional track title, artist, duration, description and the matched preferences. A repeat prompt allows another search.

Transparent selection rule

Genre is a required filter. A matching mood contributes two points, and a matching length contributes one. The first track in catalogue order wins an equal score. Mood therefore outranks length. The explanation prints only matches actually achieved; an absent genre in a future edited catalogue has a safe fallback.

Implementation and validation

C++17 uses a Track struct, a vector catalogue, loops and conditional scoring. readChoice consumes the entire line and uses istringstream to accept only one bounded integer with optional surrounding whitespace. It rejects text, decimals, trailing characters, blank input, overflow and out-of-range choices. End-of-input exits cleanly. Thirty-five black-box checks exercise 18 preference combinations and the key invalid-input, EOF, repeat and fallback cases.

Limits

Tracks and artists are invented, the catalogue is small, and weights were chosen for a classroom demonstration. No machine learning, personalised history, account, persistence, streaming, or usability study is implemented. Tests establish deterministic behaviour, not satisfaction with the selected music.

Build: g++ -std=c++17 -Wall -Wextra -pedantic src/main.cpp -o music_assistant

Part 2 Input validation source

The image is generated from the actual C++ file. readChoice validates each whole line before accepting a bounded integer. Full source is included in src/main.cpp.

```
33 int readChoice(const std::string& prompt, int minimum, int maximum) {
34     while (true) {
35         std::cout << prompt;
36         std::string line;
37         if (!std::getline(std::cin, line)) {
38             std::cout << "\nInput ended. Goodbye.\n";
39             std::exit(0);
40         }
41         std::istringstream input(line);
42         int choice = 0;
43         char extra = '\0';
44         if (input >> choice && !(input >> extra)
45             && choice >= minimum && choice <= maximum) {
46             return choice;
47         }
48         std::cout << "Please enter a whole number from " << minimum
49             << " to " << maximum << ".\n";
50     }
51 }
```

Figure 1. Actual readChoice source with line numbers.

Part 2 Recommendation source

A required genre filter, mood weight of two and length weight of one determine each result. Equal scores retain the first track; a missing-genre guard prevents a null dereference.

```
67     int bestScore = -1;
68     const Track* best = nullptr;
69     for (const Track& track : catalogue) {
70         if (track.genre != genres[genre - 1]) continue;
71         int score = 0;
72         if (track.mood == moods[mood - 1]) score += 2;
73         if ((track.minutes <= 4) == (length == 1)) score += 1;
74         // On equal scores retain the first track in catalogue order.
75         if (score > bestScore) {
76             best = &track;
77             bestScore = score;
78         }
79     }
80     // Guard against a future catalogue without the requested genre.
81     if (best == nullptr) {
82         std::cout << "No tracks are available in this genre.\n";
83         again = readChoice("Find another? 1 Yes 2 No: ", 1, 2) == 1;
84         continue;
85     }
```

Figure 2. Actual selection code and fallback.

Part 2 Programme output and validation

A genuine terminal session recommends Slow Orbit, then City Lanterns after a second search. The explanation does not claim a Reflective mood match for City Lanterns.

```
MUSIC DISCOVERY ASSISTANT
An offline, rule-based demonstration of digital music discovery.
Every track and artist is fictional; no account or internet is needed.

Genre: 1 Pop  2 Electronic  3 Acoustic
Choose genre (1-3): 2
Mood: 1 Calm  2 Energetic  3 Reflective
Choose mood (1-3): 1
Length: 1 Up to 4 min  2 Over 4 min: 2

Recommendation: Slow Orbit by Rin Echo (6 min)
A spacious instrumental with patient textures.
Why: It matches your Electronic choice, Calm mood, and length preference.
Find another? 1 Yes  2 No: 1

Genre: 1 Pop  2 Electronic  3 Acoustic
Choose genre (1-3): 1
Mood: 1 Calm  2 Energetic  3 Reflective
Choose mood (1-3): 3
Length: 1 Up to 4 min  2 Over 4 min: 1

Recommendation: City Lanterns by Mira Vale (3 min)
Soft vocals for a quiet evening.
Why: It matches your Pop choice, and length preference.
Find another? 1 Yes  2 No: 2
Thanks for exploring.
```

Figure 3. Interactive output and typed choices; complete transcript in assets/program_output.txt.

```
MUSIC DISCOVERY ASSISTANT
An offline, rule-based demonstration of digital music discovery.
Every track and artist is fictional; no account or internet is needed.

Genre: 1 Pop  2 Electronic  3 Acoustic
Choose genre (1-3): 1abc
Please enter a whole number from 1 to 3.
Choose genre (1-3): 1.5
Please enter a whole number from 1 to 3.
Choose genre (1-3): 1
```

Figure 4. Real rejection of 1abc and 1.5; complete transcript in assets/invalid_output.txt.

Part 2 Testing and results

35 black-box checks passed: 18 preference combinations, 8 invalid inputs, 4 EOF cases, repeat, whitespace, menu bounds, fallback and mood priority.

The Python driver executes the compiled programme as a separate process and writes all 35 individual outcomes to `assets/test_results.csv`. These checks validate deterministic behaviour rather than listener satisfaction.

Case	Input	Expected	Actual	Status
C01	1, 1, 1; exit 2	City Lanterns	City Lanterns	PASS
C08	2, 1, 2; exit 2	Slow Orbit	Slow Orbit	PASS
C17	3, 3, 1; exit 2	Paper Harbour	Paper Harbour	PASS
I01	'wrong'; then 1,3,1,2	Reject and recover	Rejected and recovered	PASS
I05	'1abc'; then 1,3,1,2	Reject and recover	Rejected and recovered	PASS
I06	'1.5'; then 1,3,1,2	Reject and recover	Rejected and recovered	PASS
I07	"; then 1,3,1,2	Reject and recover	Rejected and recovered	PASS
E01	"	Clean EOF exit	Clean EOF exit	PASS
R01	2,1,2,1; 1,2,1,2	Two recommendations and normal exit	Two recommendations and normal exit	PASS
B01	invalid length 3 and repeat 7	Reject at both menus and recover	Rejected at both menus	PASS
F01	Pop, Reflective, short	Fallback does not claim a mood match	No false mood match	PASS
P01	Acoustic, Energetic, short	Mood priority; no false length claim	Storm Window; no false length claim	PASS

Run `sh test.sh`. Python is needed to reproduce tests, not to run the C++ programme.

Part 2 Git development record

This screenshot is generated from the actual Git repository. Earlier Netflix commits remain in its history; subsequent Spotify work records the progressive team development stages.

```
* a8d8817 Rebuild Spotify Clayton poster report and program evidence
* a3a6704 Stage Spotify music assistant pending Topic 6A graph review
* 73ddf6a Align Netflix graph and customer mapping with supplied Clayton
template
* 620b5d1 Package reviewed report poster and reproducible evidence
* ff2cf6b Document verified Netflix lifecycle model and six-member submission
materials
* 27ea971 Validate complete menu lines and verify recommendation and recovery
scenarios
* a4ee86f Clarify foothold stage and add presentation guidance and disclosure
* 41caafc Lay out A3 poster and report with readable code capture
* f31ad38 Capture actual program run and local development history
* af73f2b Handle invalid and ended input; add five scenario checks
* 0270ef4 Implement menu inputs and fictional film catalogue
* eddae02 Document scope and learning goals for the movie assistant
```

Figure 5. Real git log --oneline --graph --all showing the complete project development timeline.

Each member should review the work and add their real contribution record. The assignment's --online example is a typo; the Git option used here is --oneline.

Source code and Git history access link: [INSERT WORKING SOURCE CODE AND GIT HISTORY URL]

Presentation links and declaration notes

OneDrive video recording: [INSERT WORKING ONEDRIVE VIDEO URL]

Source code and Git history: [INSERT WORKING SOURCE CODE AND GIT HISTORY URL]

Turnitin reports: [ATTACH REQUIRED REPORTS OR INSERT ACCESS LINK]

Speaker	Suggested subject	Approximate time
Aziel Tan Zheng Chuan	Case and thesis	1:40
See Wing Kit	Graph and customer levels	2:20
Soo Kian Rong	Timeline and market impact	2:15
Vincent Lock Chun Kit	Programme design	2:25
Wong Kee Yuan	Demo and validation	3:00
Ho Ming Hao	Testing, critique and close	2:30

The six-person script is supplied separately. The suggested total is 14:10, allowing transitions within the 18-minute maximum. The group must make its own MPEG, MP4 or MOV recording.

Submission declarations and verification

Each team member has contributed to the research, analysis, C++ program design, testing and documentation as specified in the individual contribution declarations. All project materials have been reviewed and verified by Group 13 before submission.

Each member must state their actual work and complete their own declaration. Confirm the class section, links, video, Turnitin reports and course originality requirements before submission.